

Tutorial Quiz 6 Rubric

Q1] Find the length of the curve

$$\mathbf{r}(t) = (\cos t + t \sin t) \hat{i} + (\sin t + t \cos t) \hat{j},$$

$$\sqrt{2} \leq t \leq 2$$

velocity $\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}$

$$= (-\sin t + \sin t + t \cos t) \hat{i} + (\cos t - \cos t + t \sin t) \hat{j}$$
$$= (t \cos t) \hat{i} + (t \sin t) \hat{j}$$

speed $s = |\mathbf{v}(t)|$

$$= \sqrt{t^2 \cos^2 t + t^2 \sin^2 t}$$
$$= \sqrt{t^2 (\cos^2 t + \sin^2 t)}$$
$$= |t|$$
$$= t \quad (\text{since } t > 0)$$

$\therefore$ arc length $L = \int_s db = \int_{\sqrt{2}}^2 t \, dt$

$$= \left[\frac{t^2}{2} \right]_{\sqrt{2}}^2 = \frac{4}{2} - \frac{2}{2}$$

$$= 2 - 1 = 1$$

→ ①

Q₂]

$$r(t) = (3\sin t)\hat{i} + (3\cos t)\hat{j} + 4t\hat{k}$$

$$\text{velocity } v(t) = \frac{dr}{dt} = (3\cos t)\hat{i} + (-3\sin t)\hat{j} + 4\hat{k}$$

$$\text{magnitude } |v(t)| = \sqrt{9\cos^2 t + 9\sin^2 t + 16}$$

$$= \sqrt{9 + 16} = \sqrt{25} = 5$$

∴ unit tangent

$$\text{vector } T(t) = \frac{v(t)}{|v(t)|}$$

$$= \frac{3}{5}\cos t\hat{i} - \frac{3}{5}\sin t\hat{j} + \frac{4}{5}\hat{k}$$

→ ①

$$\text{Now } \frac{dT}{dt} = -\frac{3}{5}\sin t\hat{i} - \frac{3}{5}\cos t\hat{j}$$

$$\text{and } \left| \frac{dT}{dt} \right| = \frac{3}{5}$$

$$\therefore \text{Curvature } K = \frac{1}{|v|} \cdot \left| \frac{dT}{dt} \right|$$

$$= \frac{1}{5} \cdot \frac{3}{5} = \frac{3}{25}$$

→ ①